AENL226: Power electronics and systems

Tutorial: 1

1. A 20 MVA, 11 kV synchronous generator has X_d''=18% on its own base. It feeds a 25 MVA, 11/132 kV step-up transformer with 10% leakage, followed by a 132 kV transmission line of Z=30+j120 Ω (per phase). Take the system base as 25 MVA and 132 kV on the line side. Determine the per-unit reactance/impedance of the generator (on the common base), transformer, and line on the stated base.
2. A 132/33 kV, 20 MVA transformer has 8% leakage reactance and supplies a 33/0.415 kV, 10 MVA transformer with 6% leakage reactance feeding a 0.415 kV bus. Choose a common system base of 15 MVA referred to the 0.415 kV bus. Express both transformer reactance in per-unit on the 15 MVA base. Also compute their equivalent series reactance in ohms at 0.415 kV.
3. A 50 MVA, 13.8 kV generator has X_d=0.20 p.u. on its own base and supplies a 13.8/220 kV, 63 MVA transformer (10%), a 220 kV line with Z=40+j160 Ω per phase, and a 220/33 kV, 40 MVA transformer (12%) to a 33 kV load of 30 MVA at 0.9 pf lag. Select a 50 MVA system base with voltage base 33 kV at the load bus. Compute all series elements and the load in per-unit on the chosen base.
4. A 6.6/0.415 kV distribution transformer is tested from the LV side and shows a series impedance magnitude of 0.006 Ω (referred to LV). Determine the per-unit impedance on (a) 5 MVA and (b) 10 MVA bases at 0.415 kV. Comment on how per-unit value changes with base MVA.
5. Two 33/11 kV transformers operate in parallel feeding an 11 kV bus:
   - T1: 25 MVA, 8% leakage
   - T2: 16 MVA, 6% leakage

   Adopt a common 20 MVA base at 11 kV for the bus. Find the per-unit reactance of each transformer on the 20 MVA base and their equivalent per-unit reactance in parallel.
6. A 110 kV three-phase overhead line has per-phase series impedance 0.25+j0.95 Ω/km. The line length is 60 km. Select a 100 MVA, 110 kV base. Compute the total line impedance in per-unit on this base.